

Context →	Attributes are placed within the tag itself, making additional alterations to the element content between the start and end tag. They never stand alone. They are written in the format <code>name=value</code>, where <code>name</code> is the name of the attribute (for instance <code>color</code>), and <code>value</code> describes this specific instance (for instance <code>red</code>). You've actually seen attributes before, if you followed the tutorial in the basic HTML section. <code></code> tags use the <code>src</code> attribute, anchors use the <code>name</code> attribute, and links use the <code>href</code> attribute. See how those all follow the <code>__=__</code> format? Making a table, or chart, requires several different tags. Play with these tags, or learn about HTML tables in more detail. Start with table tags around the entire table: <code><table></table></code> Row tags around the contents of each row: <code><tr></code> Column headers in the first row: <code><th></code> Cells in subsequent rows: <code><td></code> Here's an example of how it all fits together: <code><table><tr><th>Column 1: Month</th><th>Column 2: Money Saved</th></tr><tr><td>January</td><td>\$100</td></tr></table></code> You've already learned the <code><head></code> tag, which shows up at the start of each document. Besides the <code><title></code> tag, it can include the following types of tags: Meta tags, which are used to provide metadata about a web page. This data can be used by search engines when the robot scours the internet to locate and list websites. To make your website more visible on search engines, use one or more <code><meta></code> start tags (no end tags necessary), each with exactly one name attribute and one content attribute, for example: <code><meta name=description content=Write a description here></code>; or <code><meta name=keywords content=Write a list of keywords, each separated by a comma></code> <code><link></code> tags are used to associate other files with the page. This is mainly used to link to CSS stylesheets, which are made using a different type of coding to alter your HTML page by adding color, aligning your text, and many other things. <code><script></code> tags are used to link the page to JavaScript files, which can cause the page to change as... TL;DR in English:
Target →	Learn about attributes. Experiment with HTML tables. Learn the miscellaneous head tags. Play around with HTML found on websites. Learn more advanced web design from comprehensive guides.

Figure 38: Formatted dataset example from WikiLingua evaluated using ROUGE as described in Appendix K.

Context →	John travelled to the kitchen. Sandra moved to the kitchen. Daniel went to the kitchen. John journeyed to the hallway. Mary journeyed to the bedroom. Mary journeyed to the kitchen. Mary travelled to the bedroom. Sandra travelled to the bedroom. John went to the office. John went back to the kitchen. Where is Mary?
Target →	bedroom

Figure 39: Formatted dataset example from bAbI evaluated using exact match as described in Appendix K.

U Other Experiments

UL2 We experimented with the UL2 objective [106, 107] for a causal model but did not find it to outperform regular causal language modeling on our evaluation tasks. This may stem from UL2 being better suited as an Encoder-Decoder model or from mistakes in our UL2 implementation.

The Pile We have also trained several models on The Pile [31] and found similar trends as for OSCAR and C4. We make these models publicly available.

V Release of Artifacts

We open-source all of our models and code under Apache 2.0 licenses. Our filtered datasets are released with the same licenses as the datasets they stem from. All material can be found at: <https://github.com/huggingface/datablations>.

W Version Control

V4 → V5:

- Small writing improvements
- Fixed R^2 scores in Table 1

V3 → V4:

- Added comparison of different fits in terms of loss and R^2 in Table 1
- Small writing improvements

V2 → V3:

- Added loss curves of complementary strategies in Appendix P
- Fixed OSCAR validation plot in Appendix I
- Clarified the usage of smoothing in training and validation plots in Appendix K
- Added more references

V1 → V2:

- Added experiments decaying alpha and beta to allow excess epochs or parameters to hurt in Appendix F
- Added Galactica case study in Appendix G
- Added more details on the calculation of U_N given U_D in Appendix A
- Added hyperparameter sensitivity limitation in Appendix Q
- Added more detail on how score normalization is done in Appendix K
- Mentioned modification of number of heads in Appendix S

X Broader Impacts

Large language models carry potential risks such as outputting offensive language, propagating social biases, and leaking private information [119, 8]. By publicly releasing all of our models and providing new insights to improve the scaling of LLMs we may contribute to the further proliferation of these harms. However, we note that there are already much larger and more capable models freely available [14, 13, 95, 11] that can be used in such harmful ways. Thus, we consider the open-source release of our models and research to significantly outweigh its downsides.